

LOKSHAK, Russian Federation

WMO: 312630

Lat: 54.733N

Lon: 130.450E

Elev: 463

StdP: 95.88

Time Zone: 9.00 (E09)

Period: 86-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.3	-41.8	-46.3	0.0	-44.0	-44.1	0.1	-41.7	3.5	-23.9	3.2	-24.8	0.2	N/A	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.9	21.6	17.3	13.1	20.9	16.2	12.2	20.2	59.6	25.9	56.3	24.5	53.2	23.1	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.5	3.5	2.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.0	-29.8	-23.8	-14.2	-2.1	7.0	14.5	17.4	15.2	7.4	-3.3	-20.4	-29.7
	DBStd	17.83	6.14	5.22	5.86	5.12	4.51	3.78	3.36	3.13	4.08	5.67	6.97	5.59
	HDD10.0	6069	1235	946	749	363	115	7	0	3	98	412	911	1229
	HDD18.3	8573	1493	1179	1007	613	350	122	58	103	329	671	1161	1488
	CDD10.0	581	0	0	0	1	23	143	231	164	19	0	0	0
	CDD18.3	44	0	0	0	0	0	8	30	7	0	0	0	0
	CDH23.3	776	0	0	0	0	23	218	391	142	2	0	0	0
	CDH26.7	177	0	0	0	0	4	39	112	22	0	0	0	0
	WSAvg	0.6	0.4	0.5	0.8	0.9	1.0	0.8	0.7	0.6	0.6	0.6	0.5	0.3
Precipitation	PrecAvg	685	11	10	18	36	68	88	139	126	85	54	32	17
	PrecMax	848	30	28	39	78	137	149	249	192	143	113	61	41
	PrecMin	493	2	1	1	12	31	33	57	36	45	27	11	4
	PrecStd	93	7	7	10	16	27	26	56	35	25	19	14	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.6	-5.5	5.4	17.5	26.8	29.9	31.6	29.0	23.0	13.0	-2.2	-11.8
		MCWB	-11.6	-6.9	0.6	7.9	15.8	18.7	19.8	19.6	14.9	6.3	-3.9	-12.6
	2%	DB	-13.3	-8.8	1.7	11.6	22.3	27.9	29.5	26.6	20.4	9.5	-4.5	-14.8
		MCWB	-14.3	-10.3	-1.5	4.7	12.3	17.0	18.8	18.1	13.9	4.2	-6.0	-15.1
	5%	DB	-16.6	-11.8	-0.9	8.9	19.4	26.2	27.5	24.6	17.8	6.7	-6.8	-18.0
		MCWB	-17.3	-13.2	-3.6	2.9	11.1	16.0	18.3	17.6	11.9	2.6	-7.8	-18.6
	10%	DB	-20.2	-14.4	-3.5	6.5	16.6	24.2	25.2	22.7	15.4	4.6	-9.3	-20.5
		MCWB	-20.7	-15.4	-5.8	1.4	9.2	15.5	17.8	16.6	10.7	1.6	-10.1	-20.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.6	-6.7	0.9	8.2	17.1	20.8	21.2	21.5	17.0	6.6	-3.5	-12.4
		MCDB	-10.6	-5.6	4.5	16.9	24.8	26.7	27.1	25.8	20.7	11.8	-2.4	-11.9
	2%	WB	-14.2	-10.3	-1.1	5.1	13.9	18.9	20.2	19.7	15.2	4.8	-5.8	-15.3
		MCDB	-13.1	-8.9	1.5	11.2	20.2	24.2	26.5	24.3	19.3	8.1	-4.7	-14.8
	5%	WB	-17.4	-13.3	-3.6	3.3	11.8	17.6	19.3	18.6	12.8	3.2	-7.8	-18.5
		MCDB	-16.7	-12.0	-1.1	8.2	18.0	23.2	25.3	22.8	15.8	6.3	-6.8	-18.0
	10%	WB	-20.6	-15.4	-5.7	1.7	10.0	16.5	18.5	17.5	10.9	1.8	-10.1	-20.8
		MCDB	-20.1	-14.3	-3.5	5.7	16.1	22.1	23.8	21.2	15.0	4.4	-9.3	-20.5
Mean Daily Temperature Range	5% DB	MDBR	13.1	16.5	16.8	13.2	13.6	15.5	12.9	13.2	13.4	11.9	12.8	11.8
		MCDBR	12.0	17.5	16.9	16.5	19.1	19.5	17.6	17.3	17.3	14.0	11.2	12.0
		MCWBR	11.3	15.9	13.4	10.3	10.1	9.5	8.1	9.5	10.4	9.1	9.9	11.2
	5% WB	MCDBR	12.0	17.3	16.3	14.9	17.4	15.5	15.0	14.7	14.9	13.0	10.5	11.7
		MCWBR	11.3	15.8	12.8	9.6	9.6	8.7	7.8	9.4	10.2	8.7	9.4	11.2
Clear-Sky Solar Irradiance	taub		0.236	0.269	0.318	0.442	0.486	0.478	0.475	0.405	0.355	0.335	0.266	0.230
	taud		2.148	2.118	2.026	1.842	1.933	2.058	2.109	2.314	2.389	2.222	2.109	2.089
	Ebn at Noon		715	811	839	768	760	775	768	809	807	715	645	623
	Edh at Noon		61	91	128	179	174	155	145	112	90	83	63	51
All-Sky Solar Radiation	RadAvg		1.12	2.31	4.05	5.42	5.12	5.67	5.02	4.17	3.05	2.00	1.23	0.82
	RadStd		0.04	0.08	0.17	0.32	0.46	0.43	0.52	0.39	0.29	0.18	0.08	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page